

Stage 2: Production-Grade Database & ERD Specification

Alight International Multi-Vendor Marketplace Document Version: 2.0.0 (Production-Ready) Target Engine: PostgreSQL 16+ / PostgreSQL 18 Migration Framework: Flyway (`v1__initial_schema.sql` through `v30__production_erd_optimizations_and_triggers.sql`) Total Tables: 71 Base Tables + 6 System Views/Metadata

1. Architectural Principles & Enterprise Standards

- 1. **Primary Key Standard:** All business tables use cryptographically secure 128-bit UUIDs (`gen_random_uuid()`) to prevent enumeration attacks, enable client-side UUID pre-generation, and support distributed multi-warehouse synchronization.
- 2. **Optimistic Locking:** Critical transactional entities (`warehouse_stock` , `vendor_wallets` , `orders` , `vendor_orders` , `products` , `carts` , `coupons`) feature a `version BIGINT DEFAULT 0` column checked via JPA `@Version` to eliminate race conditions during high-volume flash sales and concurrent checkouts.
- 3. **Automated Timestamp Triggers:** High-performance PL/pgSQL trigger (`trg_set_updated_at`) automatically updates `updated_at` on every row mutation, eliminating application-side timestamp drift.
- 4. **Trigram & Full-Text Search:** PostgreSQL extension `pg_trgm` powers GIN indexes across product titles, SKUs, brand names, and vendor store names for sub-10ms autocomplete, typo tolerance, and faceted catalog search.
- 5. **Multi-Vendor Partitioning:** Orders are architected with a Parent-Child pattern (`orders` → `vendor_orders` → `order_items`) enabling independent vendor shipping, partial fulfillment, vendor-isolated settlements, and RMA handling.

2. Global Entity Table Dictionary (71 Tables)

Domain	Table Name	PK Type	Description	Concurrency
System	system_metadata	VARCHAR(64)	System-level health markers and migration flags	-
User & Auth	users	UUID	Customer, vendor, and administrator credentials	updated_at
User & Auth	roles	UUID	RBAC roles (CUSTOMER, VENDOR, ADMIN, SUPER_ADMIN)	Immutable
User & Auth	permissions	UUID	Fine-grained API and resource permissions	Immutable
User & Auth	user_roles	(user_id, role_id)	Cross-table mapping users to roles	-
User & Auth	role_permissions	(role_id, perm_id)	Cross-table mapping roles to permissions	-
User & Auth	user_addresses	UUID	Customer billing and shipping delivery addresses	updated_at
User & Auth	user_sessions	UUID	Active user sessions and device telemetry	-
User & Auth	refresh_tokens	UUID	JWT refresh tokens with rotation and revocation	-
Vendor	vendors	UUID	Multi-vendor business entity and verification state	updated_at
Vendor	vendor_business_details	UUID	Legal registration, GSTIN, PAN, and KYC details	updated_at
Vendor	vendor_pickup_addresses	UUID	Warehouse pickup locations for carrier dispatches	updated_at

Domain	Table Name	PK Type	Description	Concurrency
Vendor	vendor_documents	UUID	KYC documents, GST certificates, and identity proofs	-
Vendor	vendor_bank_accounts	UUID	Verified bank accounts for automated payout disbursements	updated_at
Catalog	categories	UUID	Hierarchical N-level product category tree	updated_at
Catalog	brands	UUID	Official manufacturer and designer brands	updated_at
Catalog	products	UUID	Core product master record with pricing and SEO	@Version
Catalog	product_variants	UUID	Size, finish, color, and dimension variations	updated_at
Catalog	product_images	UUID	High-res gallery media, CDN URLs, and alt texts	-
Catalog	product_attributes	UUID	Key-value technical specifications (Material, Grade)	-
Catalog	product_tags	UUID	Search tags and promotional taxonomy	-
Inventory	warehouses	UUID	Regional fulfillment hubs and vendor warehouses	updated_at
Inventory	warehouse_stock	UUID	SKU stock on-hand, reserved, and safety thresholds	@Version
Inventory	stock_reservations	UUID	Temporary cart/checkout stock locks with TTL	-
Inventory	stock_transactions	UUID	Immutable ledger of stock movements (IN/OUT/ADJ)	Immutable

Domain	Table Name	PK Type	Description	Concurrency
Inventory	inventory_audit_logs	UUID	Physical stock count reconciliation logs	Immutable
Pricing & Tax	currencies	UUID	Supported marketplace currencies (INR, USD, AED, EUR)	updated_at
Pricing & Tax	currency_exchange_rates	UUID	Real-time and manual FX conversion rates	updated_at
Pricing & Tax	tax_categories	UUID	Tax classes (Hardware 18%, Brass Craft 12%, etc.)	updated_at
Pricing & Tax	tax_jurisdictions	UUID	State/Regional tax rules (Inter-state IGST vs CGST/SGST)	updated_at
Pricing & Tax	tax_rules	UUID	Jurisdiction-to-tax category rate mappings	updated_at
Pricing & Tax	tax_rate_components	UUID	Tax breakdowns (CGST 9%, SGST 9%, IGST 18%)	updated_at
Pricing & Tax	product_tier_prices	UUID	B2B bulk purchase tiered discount pricing matrix	updated_at
Cart	carts	UUID	Customer and guest active shopping carts	@Version
Cart	cart_items	UUID	Multi-vendor cart line items and selections	updated_at
Orders	orders	UUID	Master customer order with grand total and payment state	@Version
Orders	vendor_orders	UUID	Split sub-orders routed to specific vendors	@Version
Orders	order_items	UUID	Granular purchased product variant line items	-

Domain	Table Name	PK Type	Description	Concurrency
Orders	order_addresses	UUID	Immutable delivery and billing address snapshots	-
Orders	order_status_history	UUID	Audit trail of all order workflow state transitions	Immutable
Payment	payment_transactions	UUID	Razorpay / Stripe payment gateway attempt logs	updated_at
Payment	payment_webhook_logs	UUID	Idempotent gateway webhook payload logs	Immutable
Settlement	vendor_wallets	UUID	Vendor available balance, pending escrow, and earnings	@Version
Settlement	wallet_transactions	UUID	Double-entry ledger of credits, debits, and holds	Immutable
Settlement	refund_transactions	UUID	Customer refund disbursements and reverse charges	-
Settlement	escrow_ledgers	UUID	Order hold periods before final vendor release	-
Logistics	shipping_carriers	UUID	Integrated courier partners (Delhivery, BlueDart, Blr)	updated_at
Logistics	shipping_pincode_zones	UUID	Regional pincode zones (Metro, Tier-1, Remote)	updated_at
Logistics	shipping_rate_rules	UUID	Weight and volumetric slab pricing matrix	updated_at
Logistics	shipment_packages	UUID	Physical dispatch boxes, AWB numbers, and dimensions	updated_at
Logistics	shipment_tracking_events	UUID	Carrier tracking checkpoint timeline	Immutable
Logistics	shipping_manifests	UUID	Bulk carrier pickup manifests and	-

Domain	Table Name	PK Type	Description	Concurrency
			handoff sheets	
Returns & RMA	rma_requests	UUID	Customer return/exchange claims (7-day window)	updated_at
Returns & RMA	rma_items	UUID	Specific items claimed with return reason and proof	-
Returns & RMA	rma_events	UUID	Return inspection, approval, and logistics lifecycle	Immutable
Returns & RMA	rma_policies	UUID	Vendor and category return window rules	updated_at
Returns & RMA	rma_inspections	UUID	Physical warehouse return grading reports	-
Promotions	coupons	UUID	Global and vendor-scoped discount vouchers	@Version
Promotions	coupon_usages	UUID	User coupon redemption tracking and limits	-
Promotions	coupon_categories	UUID	Category whitelist/blacklist restrictions	-
Promotions	promotion_rules	UUID	Dynamic pricing campaigns (e.g. Buy 2 Get 10%)	updated_at
Promotions	flash_sales	UUID	Timed marketplace flash sale events	updated_at
Promotions	flash_sale_products	UUID	Products assigned to flash sales with quotas	-
Reviews & QA	product_reviews	UUID	Customer verified purchase ratings and reviews	updated_at
Reviews & QA	review_images	UUID	Customer uploaded review photos	-
Reviews & QA	review_votes	UUID	Helpful / unhelpful upvotes on reviews	-

Domain	Table Name	PK Type	Description	Concurrency
Reviews & QA	product_questions	UUID	Pre-purchase customer Q&A threads	updated_at
Reviews & QA	product_answers	UUID	Vendor and expert verified answers	updated_at
Reviews & QA	question_votes	UUID	Upvotes on questions and answers	-
Support	support_tickets	UUID	Multi-channel customer & vendor dispute tickets	updated_at
Support	ticket_messages	UUID	Threaded discussion messages and attachments	-
Support	ticket_attachments	UUID	Uploaded dispute evidence and PDFs	-
Notifications	notification_templates	UUID	Email, SMS, and In-App notification blueprints	updated_at
Notifications	notifications	UUID	Real-time user alert inbox and read statuses	updated_at
Compliance	marketplace_commission_invoices	UUID	Monthly B2B GST tax invoices for vendor commissions	updated_at
Compliance	tax_compliance_ledgers	UUID	Section 52 TCS (1%) and Section 194-O TDS records	updated_at
Compliance	settlement_cycles	UUID	Weekly/Bi-weekly automated payout cycle batches	updated_at
Compliance	vendor_payouts	UUID	Bank NEFT/RTGS transaction disbursement batches	updated_at
Compliance	payout_batches	UUID	Multi-vendor consolidated payout disbursement files	updated_at

Domain	Table Name	PK Type	Description	Concurrency
Engagement	search_synonyms	UUID	Search keyword synonym maps (e.g. "telescopic" → "drawer channel")	-
Engagement	wishlists	UUID	Customer saved collections and moodboards	updated_at
Engagement	wishlist_items	UUID	Saved products inside customer wishlists	-
Engagement	product_bundles	UUID	Curated hardware kits (e.g., "Modular Kitchen Set")	updated_at
Engagement	product_bundle_items	UUID	Component products within a bundled hardware kit	-
Engagement	recently_viewed_products	UUID	Customer recent browsing history for recommendations	-
Engagement	quote_requests	UUID	B2B bulk architectural hardware Request For Quote (RFQ)	updated_at
Engagement	quote_items	UUID	Bill of Quantities (BOQ) line items for RFQ	-
Audit	audit_logs	UUID	Tamper-evident operational audit trail	Immutable

3. Domain ER Diagrams (Mermaid)

Domain 1: Multi-Vendor Catalog, Categories, Variants & Images

MERMAID

erDiagram

```
CATEGORIES ||--o{ CATEGORIES : "parent_id"
CATEGORIES ||--o{ PRODUCTS : "category_id"
BRANDS ||--o{ PRODUCTS : "brand_id"
VENDORS ||--o{ PRODUCTS : "vendor_id"
PRODUCTS ||--o{ PRODUCT_VARIANTS : "product_id"
PRODUCTS ||--o{ PRODUCT_IMAGES : "product_id"
PRODUCTS ||--o{ PRODUCT_ATTRIBUTES : "product_id"
PRODUCTS ||--o{ PRODUCT_TIER_PRICES : "product_id"
PRODUCTS ||--o{ PRODUCT_TAGS : "product_id"
```

PRODUCTS {

```
    UUID id PK
    UUID vendor_id FK
    UUID category_id FK
    UUID brand_id FK
    VARCHAR title
    VARCHAR slug UK
    NUMERIC base_price
    NUMERIC discount_price
    VARCHAR sku UK
    INTEGER stock_quantity
    VARCHAR status
    BIGINT version
    TIMESTAMP created_at
    TIMESTAMP updated_at
```

}

PRODUCT_VARIANTS {

```
    UUID id PK
    UUID product_id FK
    VARCHAR variant_name
    VARCHAR variant_sku UK
    NUMERIC price
    INTEGER stock_quantity
    BOOLEAN is_active
```

}

Domain 2: Multi-Warehouse Inventory & Real-Time Stock Engine

MERMAID

```
erDiagram
    WAREHOUSES ||--o{ WAREHOUSE_STOCK : "warehouse_id"
    PRODUCTS ||--o{ WAREHOUSE_STOCK : "product_id"
    PRODUCT_VARIANTS ||--o{ WAREHOUSE_STOCK : "variant_id"
    WAREHOUSE_STOCK ||--o{ STOCK_TRANSACTIONS : "stock_id"
    PRODUCTS ||--o{ STOCK_RESERVATIONS : "product_id"

    WAREHOUSE_STOCK {
        UUID id PK
        UUID warehouse_id FK
        UUID product_id FK
        UUID variant_id FK
        INTEGER quantity_on_hand
        INTEGER quantity_reserved
        INTEGER reorder_threshold
        INTEGER safety_stock
        BIGINT version
        TIMESTAMP updated_at
    }

    STOCK_RESERVATIONS {
        UUID id PK
        UUID product_id FK
        UUID variant_id FK
        UUID warehouse_id FK
        VARCHAR session_id
        INTEGER quantity
        VARCHAR status
        TIMESTAMP expires_at
    }
```

Domain 3: Multi-Vendor Order Splitting, Fulfillment & Escrow

MERMAID

```

erDiagram
    USERS ||--o{ ORDERS : "user_id"
    ORDERS ||--o{ VENDOR_ORDERS : "master_order_id"
    VENDORS ||--o{ VENDOR_ORDERS : "vendor_id"
    VENDOR_ORDERS ||--o{ ORDER_ITEMS : "vendor_order_id"
    ORDERS ||--o{ ORDER_ADDRESSES : "order_id"
    ORDERS ||--o{ ORDER_STATUS_HISTORY : "order_id"
    ORDERS ||--o{ PAYMENT_TRANSACTIONS : "order_id"
    VENDOR_ORDERS ||--o{ SHIPMENT_PACKAGES : "vendor_order_id"
    VENDORS ||--|| VENDOR_WALLETS : "vendor_id"
    VENDOR_WALLETS ||--o{ WALLET_TRANSACTIONS : "wallet_id"

    ORDERS {
        UUID id PK
        VARCHAR order_number UK
        UUID user_id FK
        NUMERIC grand_total
        VARCHAR order_status
        VARCHAR payment_status
        BIGINT version
        TIMESTAMP created_at
    }

    VENDOR_ORDERS {
        UUID id PK
        UUID master_order_id FK
        UUID vendor_id FK
        VARCHAR sub_order_number UK
        NUMERIC grand_total
        VARCHAR fulfillment_status
        BIGINT version
        TIMESTAMP created_at
    }

    VENDOR_WALLETS {
        UUID id PK
        UUID vendor_id FK
        NUMERIC available_balance
        NUMERIC pending_balance
        NUMERIC total_earnings
        BIGINT version
        TIMESTAMP updated_at
    }

```

4. Production Database Configuration

HikariCP High-Throughput Connection Pool

YAML

```
spring:
  datasource:
    hikari:
      pool-name: AlightHikariPool
      maximum-pool-size: 20
      minimum-idle: 5
      idle-timeout: 300000      # 5 minutes
      max-lifetime: 1200000    # 20 minutes
      connection-timeout: 20000 # 20 seconds
      leak-detection-threshold: 15000
      connection-test-query: SELECT 1
```

PostgreSQL Indexing Strategy

- **Primary Keys:** B-Tree Indexes on all UUID PKs.
- **Foreign Keys:** B-Tree Indexes on every FK column to eliminate table scan locks on cascades.
- **Search:** Trigram GIN indexes (`gin_trgm_ops`) on high-frequency textual filters.
- **Query Optimization:** Composite indexes ordered by (`filtering_column, sorting_column`).
- **Soft Filter Optimization:** Partial indexes (`WHERE status = 'ACTIVE'`) to keep memory footprint minimal.